

Филитов Миша

АКОС 2

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CH

```
1  #include <stdio.h>
2
3  int main() {
4      printf("Hello world!\n");
5      return 0;
6  }
```

Executor x86-64 gcc 16.2 (C, Editor #1) ✎ ✕

A ▾

☐ Wrap lines

 Libraries

 Overrides

 **Compilation**

x86-64 gcc 16.2



Compiler options...

Program returned: 0

Program stdout

Hello world!

<https://godbolt.org/z/aqYcf5r6s>

Компиляция

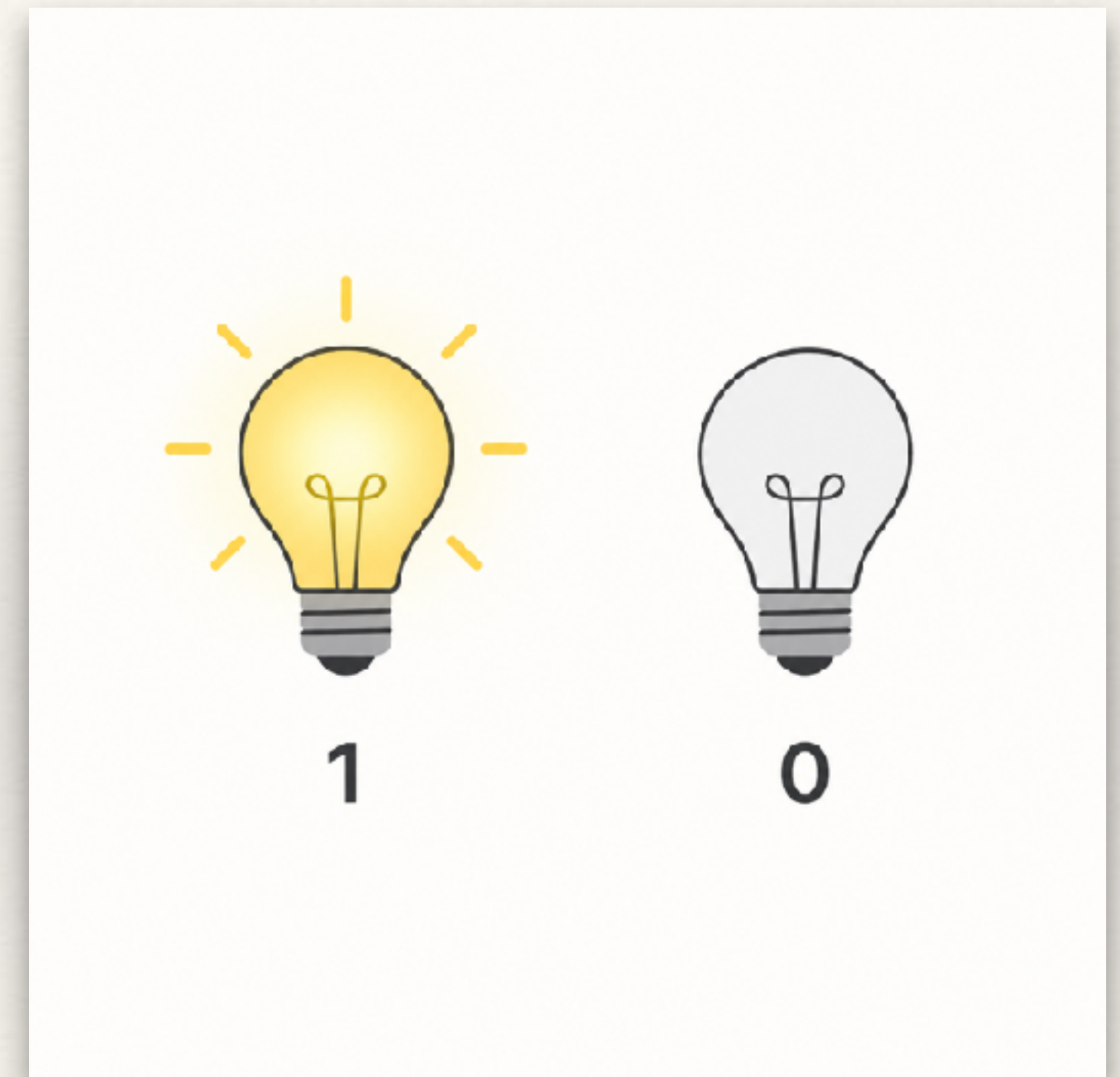
Фаза	Команда	Результат	Что делает
preprocessing	gcc -E main.c -o main.i	main.i	Обрабатывает #include, #define, #if и другие директивы
compilation	gcc -S main.i -o main.s	main.s	Переводит С-код в ассемблер
assembly	gcc -c main.s -o main.o	main.o	Переводит ассемблер в машинный код и создаёт объектный файл
linking	gcc main.o -o main	main	Объединяет объектные файлы и библиотеки в исполняемый файл

Проще

- ❖ `gcc main.c` - скомпилировали
- ❖ `./a.out` - запустили

Bit - binary digit

- ❖ Единица измерения количества информации
- ❖ Компьютеры - двоичные
- ❖ Были попытки сделать компьютеры троичными



Нестандартные байты

- ❖ БЭСМ-6 — 1 байт - 6 бит
- ❖ MIL-STD-1750A — 1 байт - 16 бит



БЭСМ-6

Байт

- ❖ В модели памяти C/C++ `byte` - минимально адресуемая единица памяти. `1 byte = 1 char`
- ❖ `sizeof(char) == 1`
- ❖ `sizeof(int) == 4`
- ❖ POSIX требует, чтобы `1 байт = 8 бит`
- ❖ `8 бит` - октет

Беззнаковые числа

$67_{10} = 64 + 2 + 1 = 2^6 + 2^1 + 2^0 = 01000011_2$

0	1	0	0	0	0	1	1
7	6	5	4	3	2	1	0

Беззнаковые числа

Dec	Bin	Hex
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7

Dec	Bin	Hex
8	1000	8
9	1001	9
10	1010	a
11	1011	b
12	1100	c
13	1101	d
14	1110	e
15	1111	f

Арифметика по модулю 2^N

$$\begin{array}{r} 1110 \\ + 1111 \\ \hline \cancel{1}1101 \end{array}$$

Dec	Bin	Hex	Dec	Bin	Hex
0	0000	0	8	1000	8
1	0001	1	9	1001	9
2	0010	2	10	1010	a
3	0011	3	11	1011	b
4	0100	4	12	1100	c
5	0101	5	13	1101	d
6	0110	6	14	1110	e
7	0111	7	15	1111	f

Переполнение легально

$$\text{UINT_MAX} + 1 = 0$$

Знаковые числа

Дополнительный код (two's complement)

Dec	Bin	Hex
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7

Dec	Bin	Hex
-8	1000	8
-7	1001	9
-6	1010	a
-5	1011	b
-4	1100	c
-3	1101	d
-2	1110	e
-1	1111	f

$$5 = 0101_2$$

$$\sim 5 = 1010_2$$

$$\sim 5 + 1 = 1011_2$$

$$-5 = 1011_2$$

Удобство дополнительного кода

Сложение / вычитание знаковых и беззнаковых чисел одинаковое

$$4 + 10 = 0b0100 + 0b1010 = 0b1110 = 14$$

$$4 + (-6) = 0b0100 + 0b1010 = 0b1110 = -2$$

$$15 + 10 = 0b1111 + 0b1010 = 0b1001 = 9$$

$$-1 + (-6) = 0b1111 + 0b1010 = 0b1001 = -7$$

Bin	Бз	З	Bin	Бз	З
0000	0	0	1000	8	-8
0001	1	1	1001	9	-7
0010	2	2	1010	10	-6
0011	3	3	1011	11	-5
0100	4	4	1100	12	-4
0101	5	5	1101	13	-3
0110	6	6	1110	14	-2
0111	7	7	1111	15	-1

Переполнение нелегально

$$\text{INT_MAX} + 1 = \text{UB}$$

Несимметричность

Dec	Bin	Hex
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7

Dec	Bin	Hex
-8	1000	8
-7	1001	9
-6	1010	a
-5	1011	b
-4	1100	c
-3	1101	d
-2	1110	e
-1	1111	f

```
int myabs(int x) {  
    if (x < 0) return -x;  
    return x;  
}
```

Несимметричность

```
1  #include <limits.h>
2  #include <stdio.h>
3
4  int myabs(int x) {
5      if (x < 0) return -x;
6      return x;
7  }
8  int main() {
9      printf("%d\n", myabs(1));
10     printf("%d\n", myabs(-1));
11     printf("%d\n", myabs(INT_MIN));
12     return 0;
13 }
```

Executor x86-64 gcc 16.2 (C, Editor #1) ✎ ✕

A ▾ □ Wrap lines 📖 Libraries 🔧 Overrides ⚙️ Compilation >_ Arguments ➡ Stdin 🔍 Runtime tools ➡ Compiler output

x86-64 gcc 16.2 ↗️ ✔️ -fsanitize=undefined

Program returned: 0

Program stdout

```
1
1
-2147483648
```

Program stderr

```
/app/example.c:5:23: runtime error: negation of -2147483648 cannot be represented in type 'int'; cast to an unsigned type to negate this value to itself
```

<https://godbolt.org/z/6q1GKT4s1>

Ловушка char

Char может быть знаковым, а может беззнаковым (implementation defined)

```
1  #include <limits.h>
2  #include <stdio.h>
3
4  int islarger(char c1, char c2) {
5      return c1 > c2;
6  }
7
8  int main() {
9      printf("1 > -1: %d\n", islarger(1, -1));
10     return 0;
11 }
```

x86-64 gcc 16.2 -fsigned-char

x86-64 gcc 16.2 -funsigned

Program returned: 0
Program stdout
1 > -1: 1

Program returned: 0
Program stdout
1 > -1: 0

<https://godbolt.org/z/j5sTnq5Kf>

Ловушка char

Char может быть знаковым, а может беззнаковым (implementation defined)

<pre>1 #include <limits.h> 2 #include <stdio.h> 3 4 int islarger(char c1, char c2) { 5 return (unsigned char)c1 > (unsigned char)c2; 6 } 7 8 int main() { 9 printf("1 > -1: %d\n", islarger(1, -1)); 10 return 0; 11 }</pre>	x86-64 gcc 16.2 Program returned: 0 Program stdout 1 > -1: 0	x86-64 gcc 16.2 Program returned: 0 Program stdout 1 > -1: 0
---	---	---

<https://godbolt.org/z/o4639n1rM>

Структуры

```
3 struct Example {
4     char c;
5     int i;
6     long long l;
7 };
8
9 int main() {
10     // Инициализация по порядку полей
11     struct Example a = {'A', 42, 1000};
12
13     // Designated initializers
14     struct Example b = {
15         .c = 'B',
16         .l = 2000
17     };
18
19     // Зануленная структура
20     struct Example c = {};
21
22     // Неинициализированная структура
23     // Читать её поля до присваивания – UB
24     struct Example d;
25
26     printf("a = {%c, %d, %lld}\n", a.c, a.i, a.l);
27     printf("b = {%c, %d, %lld}\n", b.c, b.i, b.l);
28     printf("c = {%c, %d, %lld}\n", c.c, c.i, c.l);
29     printf("d = {%c, %d, %lld}\n", d.c, d.i, d.l);
30
31     return 0;
32 }
```

x86-64 clang 23.1.0



-fsanitize=undefined

Program returned: 0

Program stdout

a = {A, 42, 1000}

b = {B, 0, 2000}

c = {, 0, 0}

d = {0, 31492, 140735183390208}

Структуры

```
3 struct Example {
4     char c;
5     int i;
6     long long l;
7 };
8
9 typedef struct Example Example;
10
11 int main() {
12     Example a = {'A', 42, 1000};
13
14     printf("a = {%c, %d, %lld}\n", a.c, a.i, a.l);
15
16     return 0;
17 }
```

<https://godbolt.org/z/PbobjPdz4>

```
3 typedef struct Example {
4     char c;
5     int i;
6     long long l;
7 } Example;
8
9 int main() {
10     Example a = {'A', 42, 1000};
11
12     printf("a = {%c, %d, %lld}\n", a.c, a.i, a.l);
13
14     return 0;
15 }
```

<https://godbolt.org/z/so1d8G3eW>

Выравнивание

- ❖ У каждого типа есть требование выравнивания `alignof(T)`. Объект типа `T` должен начинаться с адреса, подходящего для этого выравнивания
- ❖ Поля структуры идут в памяти в порядке объявления. Перед каждым полем компилятор может вставить `padding`, чтобы поле оказалось правильно выровнено
- ❖ Каждое поле выравнивается по своему `alignment`, а не по `alignment` всей структуры
- ❖ `Alignment` структуры обычно равен максимальному `alignment` её полей
- ❖ В конце структуры может добавляться `tail padding`, чтобы `sizeof(struct)` был кратен её `alignment`

Выравнивание

```
3 struct Big {
4     char c;      // size 1, align 1
5     double d;    // size 8, align 8
6     int i;       // size 4, align 4
7 };
8
9 struct Small {
10    char c;      // size 1, align 1
11    int i;       // size 4, align 4
12    double d;    // size 8, align 8
13 };
14
15
16 int main() {
17     struct Big big;
18     struct Small small;
19
20     printf("Big:\t%d\nSmall:\t%d\n", sizeof(big), sizeof(small));
21
22     return 0;
23 }
```

Executor x86-64 clang 23.1.0 (C, Editor #1)

Wrap lines Libraries Overrides Compilation Arguments Stdin

x86-64 clang 23.1.0 -fsanitize=undefined

Program returned: 0

Program stdout

Big: 24
Small: 16

<https://godbolt.org/z/sM7zd5rde>

```
3 struct Big {
4     char c;      // offset 0,      size 1
5     // offset 1..7, padding 7 bytes
6
7     double d;    // offset 8..15, size 8
8
9     int i;       // offset 16..19, size 4
10    // offset 20..23, padding 4 bytes
11 };
12
13 struct Small {
14     char c;      // offset 0,      size 1
15     // offset 1..3, padding 3 bytes
16
17     int i;       // offset 4..7,  size 4
18
19     double d;    // offset 8..15, size 8
20 }
```

<https://godbolt.org/z/dGh8Kes5K>

Массивы

```
3  int main() {
4      // Частичная инициализация
5      // остальные элементы будут инициализированы нулём
6      int arr1[10] = {};
7
8      // Designated initializer
9      // arr2[5] == 4, остальные элементы == 0
10     int arr2[10] = {[5] = 4};
11
12     // Размер массива выводится из инициализатора
13     int arr3[] = {1, 2, 3}; // 3 элемента
14
15
16     size_t len3 = sizeof(arr3) / sizeof(arr3[0]);
17
18     for (size_t i = 0; i < len3; ++i) {
19         printf("%d\n", arr3[i]);
20     }
21
22     return 0;
23 }
```

Executor x86-64 clang 23.1.0 (C, Editor #1) ✎ ✕

A ▾ □ Wrap lines 📖 Libraries 🔧 Overrides ⚙️ Compilation >_ Arguments ➡ S

x86-64 clang 23.1.0 ↗️ ✔️ -fsanitize=undefined

Program returned: 0
Program stdout

```
1
2
3
```

<https://godbolt.org/z/4K1r4o1Pj>

Массивы

```
1  #include <stdio.h>
2
3  void f1(int a[]) {
4      printf("f1:\tsizeof(a) = %zu\n", sizeof(a));
5  }
6
7  void f2(int a[100]) {
8      printf("f2:\tsizeof(a) = %zu\n", sizeof(a));
9  }
10
11 void f3(int *a) {
12     printf("f3:\tsizeof(a) = %zu\n", sizeof(a));
13 }
14
15 int main(void) {
16     int a[100];
17
18     printf("main:\tsizeof(a) = %zu\n", sizeof(a));
19
20     f1(a);
21     f2(a);
22     f3(a);
23
24     return 0;
25 }
```

x86-64 clang 23.1.0 -fsanitize=undefined

Compiler stderr

```
<source>:4:44: warning: sizeof on array function parameter will return size of 'int *' inst
4 |     printf("f1:\tsizeof(a) = %zu\n", sizeof(a));
   |                                   ^
<source>:3:13: note: declared here
3 | void f1(int a[]) {
   |             ^
<source>:8:44: warning: sizeof on array function parameter will return size of 'int *' inst
8 |     printf("f2:\tsizeof(a) = %zu\n", sizeof(a));
   |                                   ^
<source>:7:13: note: declared here
7 | void f2(int a[100]) {
   |             ^
2 warnings generated.
```

Program returned: 0

Program stdout

```
main:    sizeof(a) = 400
f1:      sizeof(a) = 8
f2:      sizeof(a) = 8
f3:      sizeof(a) = 8
```

<https://godbolt.org/z/x66o7Kq8G>

Строки

C-строка — последовательность `char`, заканчивающаяся нулевым символом `'\0'`

```
4  int main() {
5      char str[] = "hello";
6      // то же самое, что и
7      // char str[] = {'h', 'e', 'l', 'l', 'o', '\0'};
8
9      printf("sizeof = %zu\n", sizeof(str));
10     printf("strlen = %zu\n", strlen(str));
11
12     return 0;
13 }
14
```

Executor x86-64 clang 23.1.0 (C, Editor #1) ✎ ✕

A ▾ □ Wrap lines 📖 Libraries 🔧 Overrides ⚙️ Compilation >_ A

x86-64 clang 23.1.0 ✎ ✕ 🟢 -fsanitize=undefined

Program returned: 0
Program stdout
sizeof = 6
strlen = 5

<https://godbolt.org/z/zYW4rqxz6>

```
3  size_t my_strlen(const char *s) {
4      size_t i = 0;
5
6      while (s[i] != '\0') {
7          ++i;
8      }
9
10     return i;
11 }
```

<https://godbolt.org/z/nrv8WzrhT>

Строки

```

3
4  int main() {
5      char str[] = "Very very loooooooooooooooooong string";
6
7      printf("strlen = %zu\n", strlen(str));
8      printf("%s\n", str);
9
10     str[14] = '\0';
11
12     printf("strlen = %zu\n", strlen(str));
13     printf("%s\n", str);
14
15     return 0;
16 }
17

```

Executor x86-64 clang 23.1.0 (C, Editor #1)  

 A ▾ ☐ Wrap lines  Libraries  Overrides  Compilation >_ Arguments

x86-64 clang 23.1.0   -fsanitize=undefined

Program returned: 0

Program stdout

```

strlen = 35
Very very loooooooooooooooooong string
strlen = 14
Very very looo

```

<https://godbolt.org/z/95zr9a4rq>

```

1 #include <stdio.h>
2 #include <string.h>
3
4 int main(void) {
5     char somebytes[] = "1111111111111111111111111111111111111111111111111111111111111111";
6
7     char str[] = "hello";
8     printf("%zu\n", strlen(str));
9
10    str[sizeof(str)-1] = 'A';
11    printf("%zu\n", strlen(str));
12
13    return 0;
14 }

```

Executor x86-64 clang 23.1.0 (C, Editor #1)

A ▾ □ Wrap lines 📖 Libraries ⚙️ Overrides 🔧 Compilation ➤ Arguments ↗ Stdin 🔍 Runtime tools ➡ Compiler options

x86-64 clang 23.1.0 -fsanitize=undefined

Program returned: 0
 Program stdout
 5
 76

<https://godbolt.org/z/9ao1WzsWo>

Строки

```
4 int main() {
5     // char str[] = "Very very loooooooooooooooooong string";
6     char* str = "Very very loooooooooooooooooong string";
7
8     printf("strlen = %zu\n", strlen(str));
9     printf("%s\n", str);
10
11     str[14] = '\0';
12
13     printf("strlen = %zu\n", strlen(str));
14     printf("%s\n", str);
15
16     return 0;
17 }
18
```

Executor x86-64 clang 23.1.0 (C, Editor #1)

A ☐ Wrap lines Libraries Overrides Compilation Arguments

x86-64 clang 23.1.0 -fsanitize=undefined

Program returned: 1
Program stderr
UndefinedBehaviorSanitizer:DEADLYSIGNAL
==2==ERROR: UndefinedBehaviorSanitizer: SEGV on unknown address 0x5bec085e0457
==2==The signal is caused by a WRITE memory access.
#0 0x5bec085d6916 in main /app/example.c:11:13
#1 0x701e4ba2a1c9 (/lib/x86_64-linux-gnu/libc.so.6+0x2a1c9) (BuildId: 32d...
#2 0x701e4ba2a28a in __libc_start_main (/lib/x86_64-linux-gnu/libc.so.6+0...
#3 0x5bec085aa3e4 in _start (/app/output.s+0x53e4)

==2==Register values:
rax = 0x00005bec085e0457 rbx = 0x00007fff6d9feb08 rcx = 0x0000000000000001
rdi = 0x00007fff6d9fe7c0 rsi = 0x00005bec085e0401 rbp = 0x00007fff6d9fe9e0
r8 = 0x0000000000000073 r9 = 0x0000000000000000 r10 = 0x00000000ffffffff
r12 = 0x0000000000000001 r13 = 0x0000000000000000 r14 = 0x00005bec085eb648
UndefinedBehaviorSanitizer can not provide additional info.
SUMMARY: UndefinedBehaviorSanitizer: SEGV /app/example.c:11:13 in main
==2==ABORTING

<https://godbolt.org/z/67PqznnK6>

```
3 void make_uppercase(char *s) {
4     s[0] = 'H';
5 }
6
7 int main() {
8     // МОЖНО
9     char str[] = "hello";
10    // НЕЛЬЗЯ
11    // char* str = "hello";
12
13    make_uppercase(str);
14
15    printf("%s\n", str);
16 }
```

<https://godbolt.org/z/8jEqa9cE9>

Указатели

```
1  #include <stdio.h>
2
3  int main(void) {
4      int x = 67;
5      int *p = &x;
6
7      printf("x = %d\n", x);
8      printf("&x = %p\n", &x);
9      printf("p = %p\n", p);
10     printf("*p = %d\n", *p);
11
12     return 0;
13 }
```

Executor x86-64 clang 23.1.0 (C, Editor #1) ✎ ✕

A ▾ ☐ Wrap lines  Libraries  Overrides

x86-64 clang 23.1.0   -fsanit

Program returned: 0
Program stdout
x = 67
&x = 0x7fff32395898
p = 0x7fff32395898
*p = 67

<https://godbolt.org/z/5MocP1Gdf>

```
1  #include <stdio.h>
2
3  int main() {
4      int x = 67;
5      int *p = &x;
6
7      printf("x = %d\n", x);
8
9      *p = 100;
10
11     printf("x = %d\n", x);
12
13     return 0;
14 }
```

Executor x86-64 clang 23.1.0 (C, Editor #1) ✎ ✕

A ▾ ☐ Wrap lines  Libraries  Overrides

x86-64 clang 23.1.0   -fsanit

Program returned: 0
Program stdout
x = 67
x = 100

<https://godbolt.org/z/5MocP1Gdf>

Арифметика указателей

```
2
3 int main() {
4     int arr[] = {10, 20, 30, 40};
5
6     int *p = arr;
7     // int *p = &arr[0];
8
9     printf("%d\n", *p);
10    printf("%d\n", *(p + 1));
11    printf("%d\n", *(p + 2));
12    printf("%d\n", *(p + 3));
13
14    printf("p      = %p\n", (void *)p);
15    printf("p + 1 = %p\n", (void *)(p + 1));
16    printf("p + 2 = %p\n", (void *)(p + 2));
17    printf("p + 2 = %p\n", (void *)(p + 3));
18
19    return 0;
20 }
```

x86-64 clang 23.1.0

Program returned: 0

Program stdout

```
10
20
30
40
— p      = 0x7ffd8ec97ff0
p + 1 = 0x7ffd8ec97ff4
p + 2 = 0x7ffd8ec97ff8
p + 2 = 0x7ffd8ec97ffc
```

Арифметика указателей

```
2
3 int main() {
4     int arr[] = {10, 20, 30};
5
6     printf("%d\n", arr[1]);
7     printf("%d\n", *(arr + 1));
8     printf("%d\n", 1[arr]); // *(1 + arr)
9
10
11     return 0;
12 }
```

x86-64 clang 23.1.0



Program returned: 0

Program stdout

20
20
20

<https://godbolt.org/z/Pxvr97K76>

Арифметика указателей

```
4  int main() {
5      int arr[] = {10, 20, 30, 40, 50};
6
7      int *p = &arr[1];
8      int *q = &arr[4];
9
10     ptrdiff_t distance = q - p;
11
12     printf("%td\n", distance);
13
14     printf("%p\n", (void*)p);
15     printf("%p\n", (void*)q);
16     printf("%lu\n", (void*)q - (void*)p);
17
18     return 0;
19 }
```

x86-64 clang 23.1.0

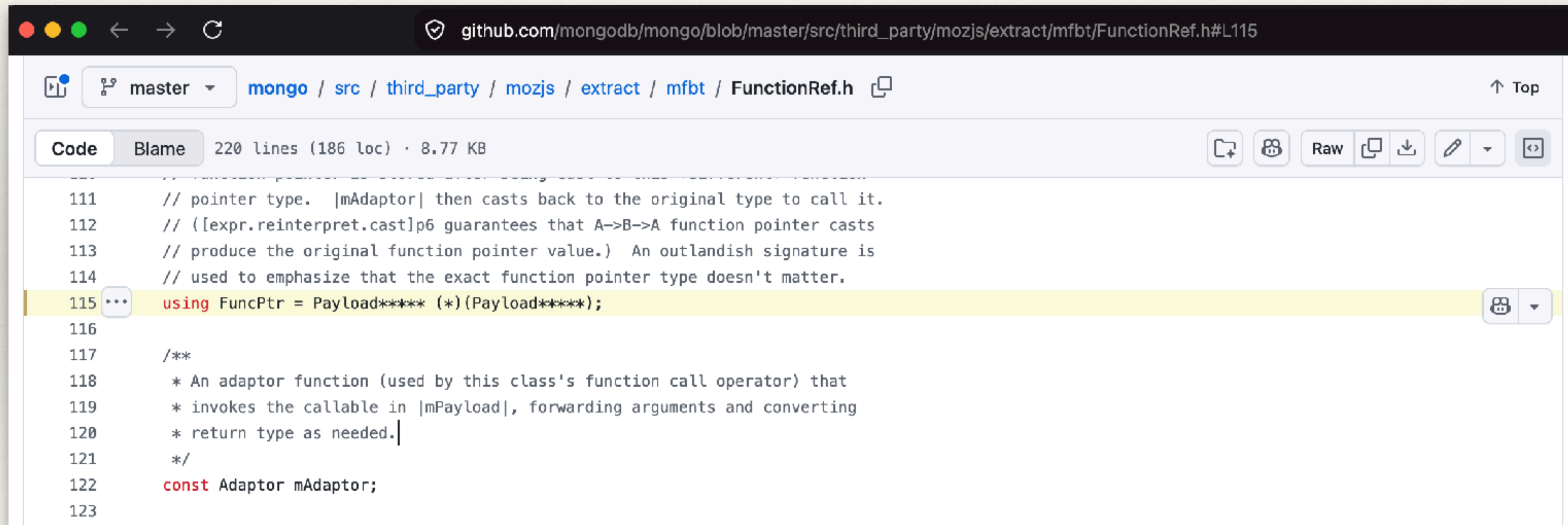
Program returned: 0

Program stdout

3
0x7ffc2b982a94
0x7ffc2b982aa0
12

<https://godbolt.org/z/Y66M4vjed>

5-star programmer



```
111 // pointer type. |mAdaptor| then casts back to the original type to call it.
112 // ([expr.reinterpret.cast]p6 guarantees that A->B->A function pointer casts
113 // produce the original function pointer value.) An outlandish signature is
114 // used to emphasize that the exact function pointer type doesn't matter.
115 ... using FuncPtr = Payload***** (*)(Payload*****);
116
117 /**
118  * An adaptor function (used by this class's function call operator) that
119  * invokes the callable in |mPayload|, forwarding arguments and converting
120  * return type as needed.
121  */
122 const Adaptor mAdaptor;
123
```

https://github.com/mongodb/mongo/blob/master/src/third_party/mozjs/extract/mfbt/FunctionRef.h#L115

Printf

```
int printf(const char *restrict format, ...);
```

```
3  int main() {
4      int age = 27;
5      double height = 1.82;
6      char grade = 'A';
7      const char *name = "Misha";
8
9      printf("age = %d\n", age);
10     printf("height = %.2f\n", height);
11     printf("grade = %c\n", grade);
12     printf("name = %s\n", name);
13
14     return 0;
15 }
```

x86-64 clang 23.1.0

Program returned: 0

Program stdout

```
age = 27
height = 1.82
grade = A
name = Misha
```

<https://godbolt.org/z/GKsdc9fY1>

Printf + while -Тьюринг полный

```
Code Blame 52 lines (47 loc) • 2.68 KB
1  #define T(a) T(a)
2  #define s(a) T(a)T(a)
3  #define A(a) s(a)T(a)a
4  #define n(a) A(a)a
5  #define D(a) n(a)A(a)
6  #define C(a) D(a)a
7  #define R C(C(N(12)G(12)))
8  #define o(a,b,c) C(H(a,a)D(G(a))C(H(b,b)G(b))n(G(b))O(32,c)R
9  #define SS O(78,55)R ""\n\033[21\n%26$s";
10 #define E(a,b,c,d) H(a,b)G(c)O(253,11)R G(11)O(255,11)R H(11,d)N(d)O(253,35)R
11 #define S(a,b) O(254,11)H(a,b)N(68)R G(68)O(255,68)N(12)H(12,68)G(67)N(67)
12
13 char* fmt = O(10,39)N(40)N(41)N(42)N(43)N(66)N(69)N(24)O(22,65)O(5,70)O(8,44)N(
14 45)N(46)N (47)N(48)N( 49)N( 50)N( 51)N(52)N(53 )O( 28,
15 54)O(5, 55) O(2, 56)O(3,57)O( 4,58 )O(13, 73)O(4,
16 71 )N( 72)O (20,59 )N(60)N(61)N( 62)N (63)N (64)R R
17 E(1,2, 3,13 )E(4, 5,6,13)E(7,8,9 ,13)E(1,4 ,7,13)E
18 {2,5,8, 13)E( 3,6,9,13)E(1,5, 9,13)E(3 ,5,7,13
19 )E(14,15, 16,23) E(17,18,19,23)E( 20, 21, 22,23)E
20 {14,17,20,23)E(15, 18,21,23)E(16,19, 22 ,23)E( 14, 18,
21 22,23)E(16,18,20, 23)R U O(255 ,38)R G ( 38)O( 255,36)
22 R H(13,23)O(255, 11)R H(11,36) O(254 ,36) R G( 36 ) O(
23 255,36)R S(1,14 )S(2,15)S(3, 16)S(4, 17 )S (5, 18)S(6,
24 19)S(7,20)S(8, 21)S(9 ,22)H(13,23 )H(36, 67 )N(11)R
25 G(11)""O(255, 25 )R s(C(G(11) ))n (G( 11) )G(
26 11)N(54)R C( "aa") s(A( G(25)))T (G(25))N (60)R o
27 {14,1,26)o( 15, 2, 27)o (16,3,28 )o( 17,4, 29)o(18
28 ,5,30)o(19 ,6,31)o( 20,7,32)o (21,8,33)o (22 ,9,
29 34)n(C(U) )N( 68)R H( 36,13)G(23) N(11)R C(D( G(11)))
30 D(G(11))G(68)N(68)R G(68)O(49,35)R H(13,23)G(67)N(11)R C(H(11,11)G(
31 11))A(G(11))C(H(36,36)G(36))s(G(36))O(32,58)R C(D(G(36)))A(G(36))SS
32
33 #define arg d+6,d+8,d+10,d+12,d+14,d+16,d+18,d+20,d+22,d+24,d+26,d+28,d+30,d+32,d+34,d+36,d+38,d+40,d+42,d+44,d+46,d+48,d+50,d+52,d+54,d+56,d+58,d+60,d+62,d+64,d+66,d+68,d+70,d+72,d+74,d+76,d+78,d+80,d+82,d+84,d+86,d+88,d+90,d+92,d+94,d+96,d+98,d+100,d+102,d+104,d+106,d+108,d+110,d+112,d+114,d+116,d+118,d+120,d+122,d+124,d+126,d+128,d+130,d+132,d+134,d+136,d+138,d+140,d+142,d+144,d+146,d+148,d+150,d+152,d+154,d+156,d+158,d+160,d+162,d+164,d+166,d+168,d+170,d+172,d+174,d+176,d+178,d+180,d+182,d+184,d+186,d+188,d+190,d+192,d+194,d+196,d+198,d+200,d+202,d+204,d+206,d+208,d+210,d+212,d+214,d+216,d+218,d+220,d+222,d+224,d+226,d+228,d+230,d+232,d+234,d+236,d+238,d+240,d+242,d+244,d+246,d+248,d+250,d+252,d+254,d+256,d+258,d+260,d+262,d+264,d+266,d+268,d+270,d+272,d+274,d+276,d+278,d+280,d+282,d+284,d+286,d+288,d+290,d+292,d+294,d+296,d+298,d+300,d+302,d+304,d+306,d+308,d+310,d+312,d+314,d+316,d+318,d+320,d+322,d+324,d+326,d+328,d+330,d+332,d+334,d+336,d+338,d+340,d+342,d+344,d+346,d+348,d+350,d+352,d+354,d+356,d+358,d+360,d+362,d+364,d+366,d+368,d+370,d+372,d+374,d+376,d+378,d+380,d+382,d+384,d+386,d+388,d+390,d+392,d+394,d+396,d+398,d+400,d+402,d+404,d+406,d+408,d+410,d+412,d+414,d+416,d+418,d+420,d+422,d+424,d+426,d+428,d+430,d+432,d+434,d+436,d+438,d+440,d+442,d+444,d+446,d+448,d+450,d+452,d+454,d+456,d+458,d+460,d+462,d+464,d+466,d+468,d+470,d+472,d+474,d+476,d+478,d+480,d+482,d+484,d+486,d+488,d+490,d+492,d+494,d+496,d+498,d+500,d+502,d+504,d+506,d+508,d+510,d+512,d+514,d+516,d+518,d+520,d+522,d+524,d+526,d+528,d+530,d+532,d+534,d+536,d+538,d+540,d+542,d+544,d+546,d+548,d+550,d+552,d+554,d+556,d+558,d+560,d+562,d+564,d+566,d+568,d+570,d+572,d+574,d+576,d+578,d+580,d+582,d+584,d+586,d+588,d+590,d+592,d+594,d+596,d+598,d+600,d+602,d+604,d+606,d+608,d+610,d+612,d+614,d+616,d+618,d+620,d+622,d+624,d+626,d+628,d+630,d+632,d+634,d+636,d+638,d+640,d+642,d+644,d+646,d+648,d+650,d+652,d+654,d+656,d+658,d+660,d+662,d+664,d+666,d+668,d+670,d+672,d+674,d+676,d+678,d+680,d+682,d+684,d+686,d+688,d+690,d+692,d+694,d+696,d+698,d+700,d+702,d+704,d+706,d+708,d+710,d+712,d+714,d+716,d+718,d+720,d+722,d+724,d+726,d+728,d+730,d+732,d+734,d+736,d+738,d+740,d+742,d+744,d+746,d+748,d+750,d+752,d+754,d+756,d+758,d+760,d+762,d+764,d+766,d+768,d+770,d+772,d+774,d+776,d+778,d+780,d+782,d+784,d+786,d+788,d+790,d+792,d+794,d+796,d+798,d+800,d+802,d+804,d+806,d+808,d+810,d+812,d+814,d+816,d+818,d+820,d+822,d+824,d+826,d+828,d+830,d+832,d+834,d+836,d+838,d+840,d+842,d+844,d+846,d+848,d+850,d+852,d+854,d+856,d+858,d+860,d+862,d+864,d+866,d+868,d+870,d+872,d+874,d+876,d+878,d+880,d+882,d+884,d+886,d+888,d+890,d+892,d+894,d+896,d+898,d+900,d+902,d+904,d+906,d+908,d+910,d+912,d+914,d+916,d+918,d+920,d+922,d+924,d+926,d+928,d+930,d+932,d+934,d+936,d+938,d+940,d+942,d+944,d+946,d+948,d+950,d+952,d+954,d+956,d+958,d+960,d+962,d+964,d+966,d+968,d+970,d+972,d+974,d+976,d+978,d+980,d+982,d+984,d+986,d+988,d+990,d+992,d+994,d+996,d+998,d+1000,d+1002,d+1004,d+1006,d+1008,d+1010,d+1012,d+1014,d+1016,d+1018,d+1020,d+1022,d+1024,d+1026,d+1028,d+1030,d+1032,d+1034,d+1036,d+1038,d+1040,d+1042,d+1044,d+1046,d+1048,d+1050,d+1052,d+1054,d+1056,d+1058,d+1060,d+1062,d+1064,d+1066,d+1068,d+1070,d+1072,d+1074,d+1076,d+1078,d+1080,d+1082,d+1084,d+1086,d+1088,d+1090,d+1092,d+1094,d+1096,d+1098,d+1100,d+1102,d+1104,d+1106,d+1108,d+1110,d+1112,d+1114,d+1116,d+1118,d+1120,d+1122,d+1124,d+1126,d+1128,d+1130,d+1132,d+1134,d+1136,d+1138,d+1140,d+1142,d+1144,d+1146,d+1148,d+1150,d+1152,d+1154,d+1156,d+1158,d+1160,d+1162,d+1164,d+1166,d+1168,d+1170,d+1172,d+1174,d+1176,d+1178,d+1180,d+1182,d+1184,d+1186,d+1188,d+1190,d+1192,d+1194,d+1196,d+1198,d+1200,d+1202,d+1204,d+1206,d+1208,d+1210,d+1212,d+1214,d+1216,d+1218,d+1220,d+1222,d+1224,d+1226,d+1228,d+1230,d+1232,d+1234,d+1236,d+1238,d+1240,d+1242,d+1244,d+1246,d+1248,d+1250,d+1252,d+1254,d+1256,d+1258,d+1260,d+1262,d+1264,d+1266,d+1268,d+1270,d+1272,d+1274,d+1276,d+1278,d+1280,d+1282,d+1284,d+1286,d+1288,d+1290,d+1292,d+1294,d+1296,d+1298,d+1300,d+1302,d+1304,d+1306,d+1308,d+1310,d+1312,d+1314,d+1316,d+1318,d+1320,d+1322,d+1324,d+1326,d+1328,d+1330,d+1332,d+1334,d+1336,d+1338,d+1340,d+1342,d+1344,d+1346,d+1348,d+1350,d+1352,d+1354,d+1356,d+1358,d+1360,d+1362,d+1364,d+1366,d+1368,d+1370,d+1372,d+1374,d+1376,d+1378,d+1380,d+1382,d+1384,d+1386,d+1388,d+1390,d+1392,d+1394,d+1396,d+1398,d+1400,d+1402,d+1404,d+1406,d+1408,d+1410,d+1412,d+1414,d+1416,d+1418,d+1420,d+1422,d+1424,d+1426,d+1428,d+1430,d+1432,d+1434,d+1436,d+1438,d+1440,d+1442,d+1444,d+1446,d+1448,d+1450,d+1452,d+1454,d+1456,d+1458,d+1460,d+1462,d+1464,d+1466,d+1468,d+1470,d+1472,d+1474,d+1476,d+1478,d+1480,d+1482,d+1484,d+1486,d+1488,d+1490,d+1492,d+1494,d+1496,d+1498,d+1500,d+1502,d+1504,d+1506,d+1508,d+1510,d+1512,d+1514,d+1516,d+1518,d+1520,d+1522,d+1524,d+1526,d+1528,d+1530,d+1532,d+1534,d+1536,d+1538,d+1540,d+1542,d+1544,d+1546,d+1548,d+1550,d+1552,d+1554,d+1556,d+1558,d+1560,d+1562,d+1564,d+1566,d+1568,d+1570,d+1572,d+1574,d+1576,d+1578,d+1580,d+1582,d+1584,d+1586,d+1588,d+1590,d+1592,d+1594,d+1596,d+1598,d+1600,d+1602,d+1604,d+1606,d+1608,d+1610,d+1612,d+1614,d+1616,d+1618,d+1620,d+1622,d+1624,d+1626,d+1628,d+1630,d+1632,d+1634,d+1636,d+1638,d+1640,d+1642,d+1644,d+1646,d+1648,d+1650,d+1652,d+1654,d+1656,d+1658,d+1660,d+1662,d+1664,d+1666,d+1668,d+1670,d+1672,d+1674,d+1676,d+1678,d+1680,d+1682,d+1684,d+1686,d+1688,d+1690,d+1692,d+1694,d+1696,d+1698,d+1700,d+1702,d+1704,d+1706,d+1708,d+1710,d+1712,d+1714,d+1716,d+1718,d+1720,d+1722,d+1724,d+1726,d+1728,d+1730,d+1732,d+1734,d+1736,d+1738,d+1740,d+1742,d+1744,d+1746,d+1748,d+1750,d+1752,d+1754,d+1756,d+1758,d+1760,d+1762,d+1764,d+1766,d+1768,d+1770,d+1772,d+1774,d+1776,d+1778,d+1780,d+1782,d+1784,d+1786,d+1788,d+1790,d+1792,d+1794,d+1796,d+1798,d+1800,d+1802,d+1804,d+1806,d+1808,d+1810,d+1812,d+1814,d+1816,d+1818,d+1820,d+1822,d+1824,d+1826,d+1828,d+1830,d+1832,d+1834,d+1836,d+1838,d+1840,d+1842,d+1844,d+1846,d+1848,d+1850,d+1852,d+1854,d+1856,d+1858,d+1860,d+1862,d+1864,d+1866,d+1868,d+1870,d+1872,d+1874,d+1876,d+1878,d+1880,d+1882,d+1884,d+1886,d+1888,d+1890,d+1892,d+1894,d+1896,d+1898,d+1900,d+1902,d+1904,d+1906,d+1908,d+1910,d+1912,d+1914,d+1916,d+1918,d+1920,d+1922,d+1924,d+1926,d+1928,d+1930,d+1932,d+1934,d+1936,d+1938,d+1940,d+1942,d+1944,d+1946,d+1948,d+1950,d+1952,d+1954,d+1956,d+1958,d+1960,d+1962,d+1964,d+1966,d+1968,d+1970,d+1972,d+1974,d+1976,d+1978,d+1980,d+1982,d+1984,d+1986,d+1988,d+1990,d+1992,d+1994,d+1996,d+1998,d+2000,d+2002,d+2004,d+2006,d+2008,d+2010,d+2012,d+2014,d+2016,d+2018,d+2020,d+2022,d+2024,d+2026,d+2028,d+2030,d+2032,d+2034,d+2036,d+2038,d+2040,d+2042,d+2044,d+2046,d+2048,d+2050,d+2052,d+2054,d+2056,d+2058,d+2060,d+2062,d+2064,d+2066,d+2068,d+2070,d+2072,d+2074,d+2076,d+2078,d+2080,d+2082,d+2084,d+2086,d+2088,d+2090,d+2092,d+2094,d+2096,d+2098,d+2100,d+2102,d+2104,d+2106,d+2108,d+2110,d+2112,d+2114,d+2116,d+2118,d+2120,d+2122,d+2124,d+2126,d+2128,d+2130,d+2132,d+2134,d+2136,d+2138,d+2140,d+2142,d+2144,d+2146,d+2148,d+2150,d+2152,d+2154,d+2156,d+2158,d+2160,d+2162,d+2164,d+2166,d+2168,d+2170,d+2172,d+2174,d+2176,d+2178,d+2180,d+2182,d+2184,d+2186,d+2188,d+2190,d+2192,d+2194,d+2196,d+2198,d+2200,d+2202,d+2204,d+2206,d+2208,d+2210,d+2212,d+2214,d+2216,d+2218,d+2220,d+2222,d+2224,d+2226,d+2228,d+2230,d+2232,d+2234,d+2236,d+2238,d+2240,d+2242,d+2244,d+2246,d+2248,d+2250,d+2252,d+2254,d+2256,d+2258,d+2260,d+2262,d+2264,d+2266,d+2268,d+2270,d+2272,d+2274,d+2276,d+2278,d+2280,d+2282,d+2284,d+2286,d+2288,d+2290,d+2292,d+2294,d+2296,d+2298,d+2300,d+2302,d+2304,d+2306,d+2308,d+2310,d+2312,d+2314,d+2316,d+2318,d+2320,d+2322,d+2324,d+2326,d+2328,d+2330,d+2332,d+2334,d+2336,d+2338,d+2340,d+2342,d+2344,d+2346,d+2348,d+2350,d+2352,d+2354,d+2356,d+2358,d+2360,d+2362,d+2364,d+2366,d+2368,d+2370,d+2372,d+2374,d+2376,d+2378,d+2380,d+2382,d+2384,d+2386,d+2388,d+2390,d+2392,d+2394,d+2396,d+2398,d+2400,d+2402,d+2404,d+2406,d+2408,d+2410,d+2412,d+2414,d+2416,d+2418,d+2420,d+2422,d+2424,d+2426,d+2428,d+2430,d+2432,d+2434,d+2436,d+2438,d+2440,d+2442,d+2444,d+2446,d+2448,d+2450,d+2452,d+2454,d+2456,d+2458,d+2460,d+2462,d+2464,d+2466,d+2468,d+2470,d+2472,d+2474,d+2476,d+2478,d+2480,d+2482,d+2484,d+2486,d+2488,d+2490,d+2492,d+2494,d+2496,d+2498,d+2500,d+2502,d+2504,d+2506,d+2508,d+2510,d+2512,d+2514,d+2516,d+2518,d+2520,d+2522,d+2524,d+2526,d+2528,d+2530,d+2532,d+2534,d+2536,d+2538,d+2540,d+2542,d+2544,d+2546,d+2548,d+2550,d+2552,d+2554,d+2556,d+2558,d+2560,d+2562,d+2564,d+2566,d+2568,d+2570,d+2572,d+2574,d+2576,d+2578,d+2580,d+2582,d+2584,d+2586,d+2588,d+2590,d+2592,d+2594,d+2596,d+2598,d+2600,d+2602,d+2604,d+2606,d+2608,d+2610,d+2612,d+2614,d+2616,d+2618,d+2620,d+2622,d+2624,d+2626,d+2628,d+2630,d+2632,d+2634,d+2636,d+2638,d+2640,d+2642,d+2644,d+2646,d+2648,d+2650,d+2652,d+2654,d+2656,d+2658,d+2660,d+2662,d+2664,d+2666,d+2668,d+2670,d+2672,d+2674,d+2676,d+2678,d+2680,d+2682,d+2684,d+2686,d+2688,d+2690,d+2692,d+2694,d+2696,d+2698,d+2700,d+2702,d+2704,d+2706,d+2708,d+2710,d+2712,d+2714,d+2716,d+2718,d+2720,d+2722,d+2724,d+2726,d+2728,d+2730,d+2732,d+2734,d+2736,d+2738,d+2740,d+2742,d+2744,d+2746,d+2748,d+2750,d+2752,d+2754,d+2756,d+2758,d+2760,d+2762,d+2764,d+2766,d+2768,d+2770,d+2772,d+2774,d+2776,d+2778,d+2780,d+2782,d+2784,d+2786,d+2788,d+2790,d+2792,d+2794,d+2796,d+2798,d+2800,d+2802,d+2804,d+2806,d+2808,d+2810,d+2812,d+2814,d+2816,d+2818,d+2820,d+2822,d+2824,d+2826,d+2828,d+2830,d+2832,d+2834,d+2836,d+2838,d+2840,d+2842,d+2844,d+2846,d+2848,d+2850,d+2852,d+2854,d+2856,d+2858,d+2860,d+2862,d+2864,d+2866,d+2868,d+2870,d+2872,d+2874,d+2876,d+2878,d+2880,d+2882,d+2884,d+2886,d+2888,d+2890,d+2892,d+2894,d+2896,d+2898,d+2900,d+2902,d+2904,d+2906,d+2908,d+2910,d+2912,d+2914,d+2916,d+2918,d+2920,d+2922,d+2924,d+2926,d+2928,d+2930,d+2932,d+2934,d+2936,d+2938,d+2940,d+2942,d+2944,d+2946,d+2948,d+2950,d+2952,d+2954,d+2956,d+2958,d+2960,d+2962,d+2964,d+2966,d+2968,d+2970,d+2972,d+2974,d+2976,d+2978,d+2980,d+2982,d+2984,d+2986,d+2988,d+2990,d+2992,d+2994,d+2996,d+2998,d+3000,d+3002,d+3004,d+3006,d+3008,d+3010,d+3012,d+3014,d+3016,d+3018,d+3020,d+3022,d+3024,d+3026,d+3028,d+3030,d+3032,d+3034,d+3036,d+3038,d+3040,d+3042,d+3044,d+3046,d+3048,d+3050,d+3052,d+3054,d+3056,d+3058,d+3060,d+3062,d+3064,d+3066,d+3068,d+3070,d+3072,d+3074,d+3076,d+3078,d+3080,d+3082,d+3084,d+3086,d+3088,d+3090,d+3092,d+3094,d+3096,d+3098,d+3100,d+3102,d+3104,d+3106,d+3108,d+3110,d+3112,d+3114,d+3116,d+3118,d+3120,d+3122,d+3124,d+3126,d+3128,d+3130,d+3132,d+3134,d+3136,d+3138,d+3140,d+3142,d+3144,d+3146,d+3148,d+3150,d+3152,d+3154,d+3156,d+3158,d+3160,d+3162,d+3164,d+3166,d+3168,d+3170,d+3172,d+3174,d+3176,d+3178,d+3180,d+3182,d+3184,d+3186,d+3188,d+3190,d+3192,d+3194,d+3196,d+3198,d+3200,d+3202,d+3204,d+3206,d+3208,d+3210,d+3212,d+3214,d+3216,d+3218,d+3220,d+3222,d+3224,d+3226,d+3228,d+3230,d+3232,d+3234,d+3236,d+3238,d+3240,d+3242,d+3244,d+3246,d+3248,d+3250,d+3252,d+3254,d+3256,d+3258,d+3260,d+3262,d+3264,d+3266,d+3268,d+3270,d+3272,d+3274,d+3276,d+3278,d+3280,d+3282,d+3284,d+3286,d+3288,d+3290,d+3292,d+3294,d+3296,d+3298,d+3300,d+3302,d+3304,d+3306,d+3308,d+3310,d+3312,d+3314,d+3316,d+3318,d+3320,d+3322,d+3324,d+3326,d+3328,d+3330,d+3332,d+3334,d+3336,d+3338,d+3340,d+3342,d+3344,d+3346,d+3348,d+3350,d+3352,d+3354,d+3356,d+3358,d+3360,d+3362,d+3364,d+3366,d+3368,d+3370,d+3372,d+3374,d+3376,d+3378,d+3380,d+3382,d+3384,d+3386,d+3388,d+3390,d+3392,d+3394,d+3396,d+3398,d+3400,d+3402,d+3404,d+3406,d+3408,d+3410,d+3412,d+3414,d+3416,d+3418,d+3420,d+3422,d+3424,d+3426,d+3428,d+3430,d+3432,d+3434,d+3436,d+3438,d+3440,d+3442,d+3444,d+3446,d+3448,d+3450,d+3452,d+3454,d+3456,d+3458,d+3460,d+3462,d+3464,d+3466,d+3468,d+3470,d+3472,d+3474,d+3476,d+3478,d+3480,d+3482,d+3484,d+3486,d+3488,d+3490,d+3492,d+3494,d+3496,d+3498,d+3500,d+3502,d+3504,d+3506,d+3508,d+3510,d+3512,d+3514,d+3516,d+3518,d+3520,d+3522,d+3524,d+3526,d+3528,d+3530,d+3532,d+3534,d+3536,d+3538,d+3540,d+3542,d+3544,d+3546,d+3548,d+3550,d+3552,d+3554,d+3556,d+3558,d+3560,d+3562,d+3564,d+3566,d+3568,d+3570,d+3572,d+3574,d+3576,d+3578,d+3580,d+3582,d+3584,d+3586,d+3588,d+3590,d+3592,d+3594,d+3596,d+3598,d+3600,d+3602,d+3604,d+3606,d+3608,d+3610,d+3612,d+3614,d+3616,d+3618,d+3620,d+3622,d+3624,d+3626,d+3628,d+3630,d+3632,d+3634,d+3636,d+3638,d+3640,d+3642,d+3644,d+3646,d+3648,d+3650,d+3652,d+3654,d+3656,d+3658,d+3660,d+3662,d+3664,d+3666,d+3668,d+3670,d+3672,d+3674,d+3676,d+3678,d+3680,d+3682,d+3684,d+3686,d+3688,d+3690,d+3692,d+3694,d+3696,d+3698,d+3700,d+3702,d+3704,d+3706,d+3708,d+3710,d+3712,d+3714,d+3716,d+3718,d+3720,d+3722,d+3724,d+3726,d+3728,d+3730,d+3732,d+3734,d+3736,d+3738,d+3740,d+3742,d+3744,d+3746,d+3748,d+3750,d+3752,d+3754,d+3756,d+3758,d+3760,d+3762,d+3764,d+3766,d+3768,d+3770,d+3772,d+3774,d+3776,d+3778,d+3780,d+3782,d+3784,d+3786,d+3788,d+3790,d+3792,d+3794,d+3796,d+3798,d+3800,d+3802,d+3804,d+3806,d+3808,d+3810,d+3812,d+3814,d+3816,d+3818,d+3820,d+3822,d+3824,d+3826,d+3828,d+3830,d+3832,d+3834,d+3836,d+3838,d+3840,d+3842,d+3844,d+3846,d+3848,d+3850,d+3852,d+3854,d+385
```

Scanf

`int scanf(const char *restrict format, ...);`

```
3  int main() {
4      int age;
5      double height;
6      char name[32];
7
8      printf("Enter age, height and name: ");
9
10     scanf("%d %lf %31s", &age, &height, name);
11
12     printf("age = %d\n", age);
13     printf("height = %.2f\n", height);
14     printf("name = %s\n", name);
15
16     return 0;
17 }
```

x86-64 clang 23.1.0

Program returned: 0

Program stdout

Enter age, height and name: age = 5
height = 6.00
name = misha

<https://godbolt.org/z/YrEjrhznv>